Salahaddin University-Erbil
College of Computer Science and IT
Stage 2 Cs & IT

Web Technology Basics - Lec. 2 - HTML

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Tuesday: IT- 8:30-10:30/ Hall 3
CS- 10:30 -12:30/ Hall 16

What you will gain from this course:

- Understand how websites work (browser ↔ server).
- Build simple, semantic pages with HTML5.
- Style and layout pages with CSS.
- Make pages responsive, accessible, and performant.

Lists & tables: presenting structured information

What Is an HTML Table?

- An HTML table is used to display data in rows and columns
- It is suitable for structured data such as schedules and results
- Tables improve data readability and organization

Why Use Tables?

- To organize related data clearly
- To display numerical or textual data systematically
- To make information easy to compare

Basic Table Tags
<table> – Defines the table
<tr> – Defines a table row
<td> – Defines a table cell

Lists & tables: presenting structured information

Basic Table Structure (one cell)

```
<table>
  <tr>
      <td>Data</td>
  </tr>
</table>
```

Data

Table Data Cells (<td>)

- <td> defines standard data cells
- Cells hold text, numbers, images, ..
- Each row can have multiple cells

Table Rows (<tr>)

- Each row is defined using the <tr> tag
- A table can have multiple rows
- Rows contain table cells

```
<table>
  <tr>
      <td>Data row 1 (cell 1)</td>
  </tr>
  <tr>
      <td>Data row 2 (cell 1)</td>
  </tr>
</table>
```

Data row 1 (cell 1)
Data row 2 (cell 1)

Lists & tables: presenting structured information

Table Headers (<th>)

- <th> defines header cells
- Header cells are bold and centre by default
- Used to describe column or row headings

Table Borders

- Borders make table structure visible
- border attribute can be used (HTML)
- CSS is recommended for styling tables

Complete Table Example

```
<table border="2" >
      <tr>
        <th>Name</th>
        <th>Age</th>
      </tr>
      <tr>
        <td>Ali</td>
        <td>20</td>
      </tr>
</table>
```

Name	Age
Ali	20

Lists & tables: presenting structured information

Cell Padding and Cell Spacing

- Cell Padding adds space inside cells
- Cell Spacing adds space between cells

Name	Age
Ali	20

Name	Age
Ali	20

Rowspan and Colspan

- rowspan merges cells vertically
- colspan merges cells horizontally
- Used for complex table layouts

rowspan

```
<table border="2“>
 <tr>
   <th>Name</th>
   <th>Age</th>
 </tr>
 <tr>
   <td>Ali</td>
   <td rowspan="2">20</td>
 </tr>
 <tr>
   <td>Ahmed</td>
 </tr>
</table>
```

Name	Age
Ali	20
Ahmed	

colspan

```
<table border="2" cellpadding="4">
<tr>
 <td colspan="2">Student Info</td>
 </tr>
 <tr>
 <td>Name</td>
 <td>Ali</td>
 </tr>
 <tr>
 <td>Age</td>
 <td>20</td>
 </tr>
 </table>
```

Student Info	
Name	Ali
Age	20

Lists & tables: presenting structured information

Header Cells: <th> and scope

- <th> tag is used to label columns/rows (e.g., “Name”, “ID”).
- Attribute **scope**="col" for column headers and **scope**="row“ for row headers.
- Helps screen readers and improves meaning (not just appearance).

Month	Product	Quantity
May	HeadPhones	20
June	CellPhone	25
July	Tabs	14

```
<table border="1" >
    <tr>
        <th scope="col">Month</th>
        <th scope="col">Product</th>
        <th scope="col">Quantity</th>
    </tr>
    <tr>
        <th scope="row">May</th>
        <td>HeadPhones</td>
        <td>20</td>
    </tr>
    <tr>
        <th scope="row">June</th>
        <td>CellPhone</td>
        <td>25</td>
    </tr>
    <tr>
        <th scope="row">July</th>
        <td>Tabs</td>
        <td>14</td>        </tr>  </table>
```

Align, Width, Height, Valign Cell Attributes

- Align: Cell Data Horizontal alignment
  align: left | centre | right
- Valign: Cell Data Vertical alignment
  valign: top | middle | bottom
- Width: adjust the width of a cell
- Height: adjust the height of a cell

Cell Attributes

Cell Attributes

```
<table border="1" cellpadding="20">
 <tr>
  <td width="80" height="90" align="right" valign="bottom“ bgcolor="yellow">align test</td>
  </tr>
 </table>
```


Table Title: <caption>

- <caption> tag provides the table's title/description (what the table is about).
- Placement rule: if used, <caption> must be the first child of <table>.

```
<table border="1" cellpadding="20">
<caption align="bottom">Student Records (Stage 1)</caption>
   <tr>
     <th>Name</th><th>ID</th><th>Dept</th>
   </tr>
   <tr>
     <td>Ali</td><td>123</td><td>CS</td>
   </tr>
 </table>
```

Student Records (Stage 1)

Name	ID	Dept
Ali	123	CS

Column Styling: <colgroup> and <col>

- <colgroup> tag: groups columns so you can style them as a unit.
- Inside it, use <col> tag for individual columns (or groups).
- span="N“ on <col> applies the same style to N **consecutive** columns.
- Placement rule: <colgroup> comes after <caption> but before <thead>/<tbody>/<tfoot>.

Table Sections: <thead>, <tbody>, <tfoot>

- <thead>: Combine rows that label the columns (headers).
- <tbody>: combine main data rows (the body of the table).
- <tfoot>: combine summary rows (totals/averages/notes).
- These sections improve readability, printing, and accessibility.

Inside <table>, order starts like this:

- <caption>
- <colgroup>
- <thead>
- <tbody>
- <tfoot>

Column Styling: <colgroup> and <col>

```
<table border="1" cellpadding="20" >
 <caption>Council budget 2018</caption>
 <colgroup>
  <col style="width:70%" bgcolor="gray">
  <col style="width:30%">
 </colgroup>
 <thead>
  <tr>
   <th scope="col" bgcolor="lightgray">Item</th>
   <th scope="col" bgcolor="lightgray">Cost</th>
  </tr>
 </thead>
 <tbody>
  <tr><th scope="row">Donuts</th><td class="num">3,000</td></tr>
  <tr><th scope="row">Stationery</th><td class="num">18,000</td></tr>
 </tbody>
 <tfoot>
  <tr><th scope="row">Total</th><td class="num">21,000</td></tr>
 </tfoot>
 </table>
```

Council budget (in £) 2018

Item	Cost
Donuts	3,000
Stationery	18,000
Total	21,000